Comsats University Islamabad, Lahore Campus

Sessional II – FALL 2020

Course Title	Information Security	Course Code		CSC432	Credit Hours		3(2,1)
Instructor/s	Dr. M Sharjeel, Dr. Atif Saeed	Program Name		BS (SE), BS (CS)			
Semester	8th	Batch	SP17	Section	A	Date	Nov 18, 2020
Time Allowed	1 hr 30 minutes	Maximum Marks		25			
Student Name:		Registration No.					

Important Instructions/Guidelines

1. Avoid writing long stories
2. Use of S-Box, Lookup table cheat sheet is allowed
3. Exchange of calculator is not allowed
4. Match of solution from a certain threshold will result in failure of entire course!
5. Return the question paper along with the answer sheet. **Missing paper will result in elimination of 5 marks**

Question no. 1 **[3]; PLO-1; CLO :<2,3>; Bloom Taxonomy Level: <*Understanding*>**

You have been hired as a cybersecurity consultant for a healthcare organization that has recently suffered a data breach. During a briefing with the executive team, they ask you to explain the fundamental concepts that guide their information strategy moving forward. Describe and explain the components contributes to the overall security of the organization's information system, particularly in protecting patient data.

Question no. 2 **[2]; PLO-2; CLO :<1>; Bloom Taxonomy Level: <*Remembering*>**

Provide security solution to the problem:

A program opens a file in write mode, but after a few hours, writing data to file fails due to permission errors.

Question no. 3 **[2*5]; PLO-3; CLO :< 3,4>; Bloom Taxonomy Level: <*Analyzing*>**

1. How Avalanche effect makes DES a better choice?
2. Illustrate the decryption process of Cipher Feedback Mode (CFB).
3. Write recursive round constant generation algorithm of AES.
4. Apply AFFINE cypher on "Sessional" with b = 8.
5. Show that in DES the first 24 bits of each sub-key come from the same subset of 28 bits of the initial key and that the second 24 bits of each sub-key come from a disjoint subset of 28 bits of the initial key.

Question no. 4 **[10]; PLO-1; CLO :< 3>; Bloom Taxonomy Level: <*Applying*>**

Start with the same bit pattern for the key and the plaintext, namely:

Binary Stream: 1010 1011 1010 1011 1100 0111 1110 0111 1000 1001 1010 1011 1100 1101 1110 1111

a. Derive the first-round sub-key.

b. Derive L_0, R_0

c. Expand R_0 to get $E[R_0]$

d. Calculate $A = E[R_0] \text{ XOR } K_1$

e. Group the 48-bit result of (d) into sets of 6 bits and evaluate the corresponding S-box substitutions.

f. Concatenate the results of (e) to get a 32-bit result

S_1	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S_2	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_3	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S_4	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S_5	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S_6	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S_7	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11